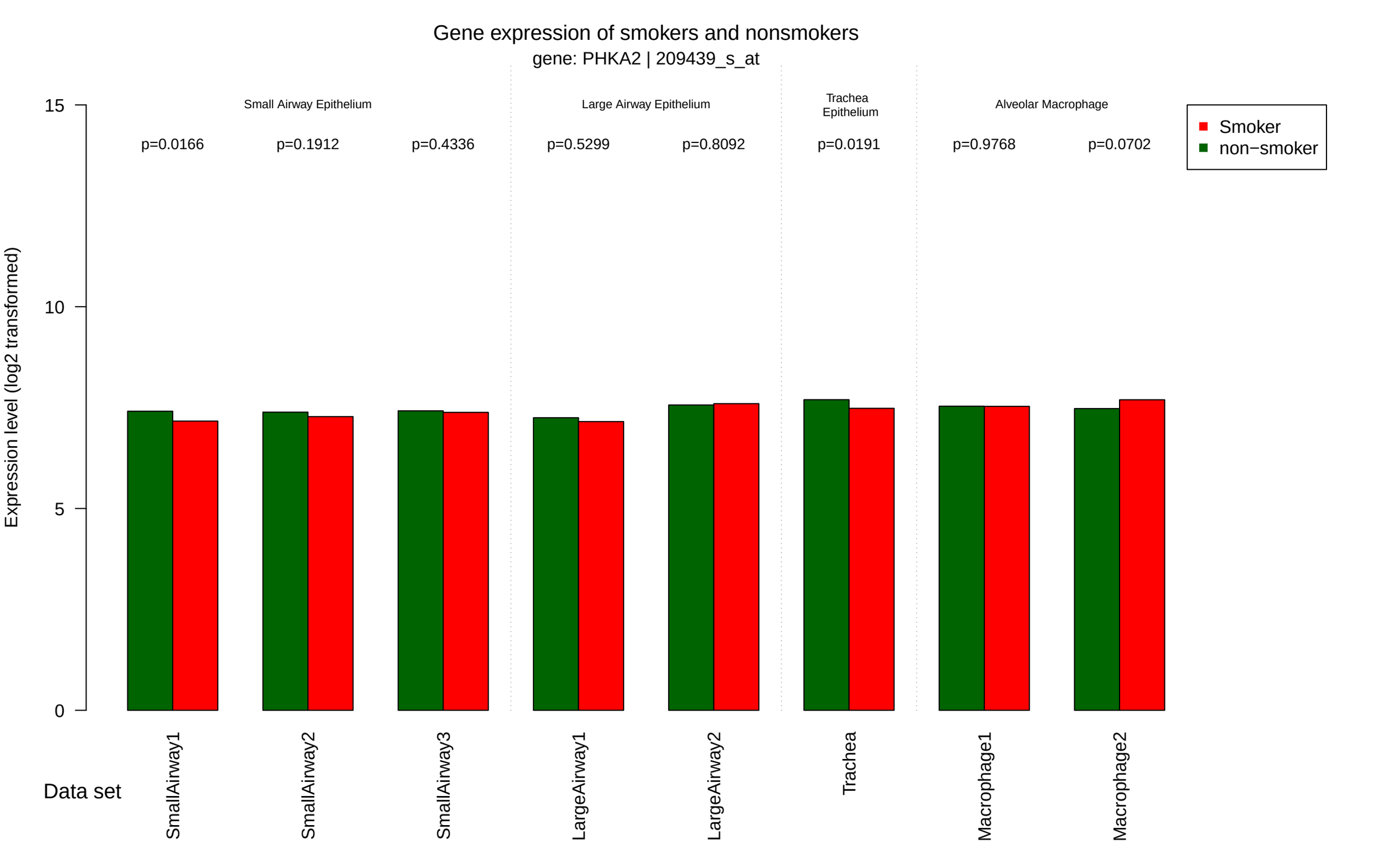
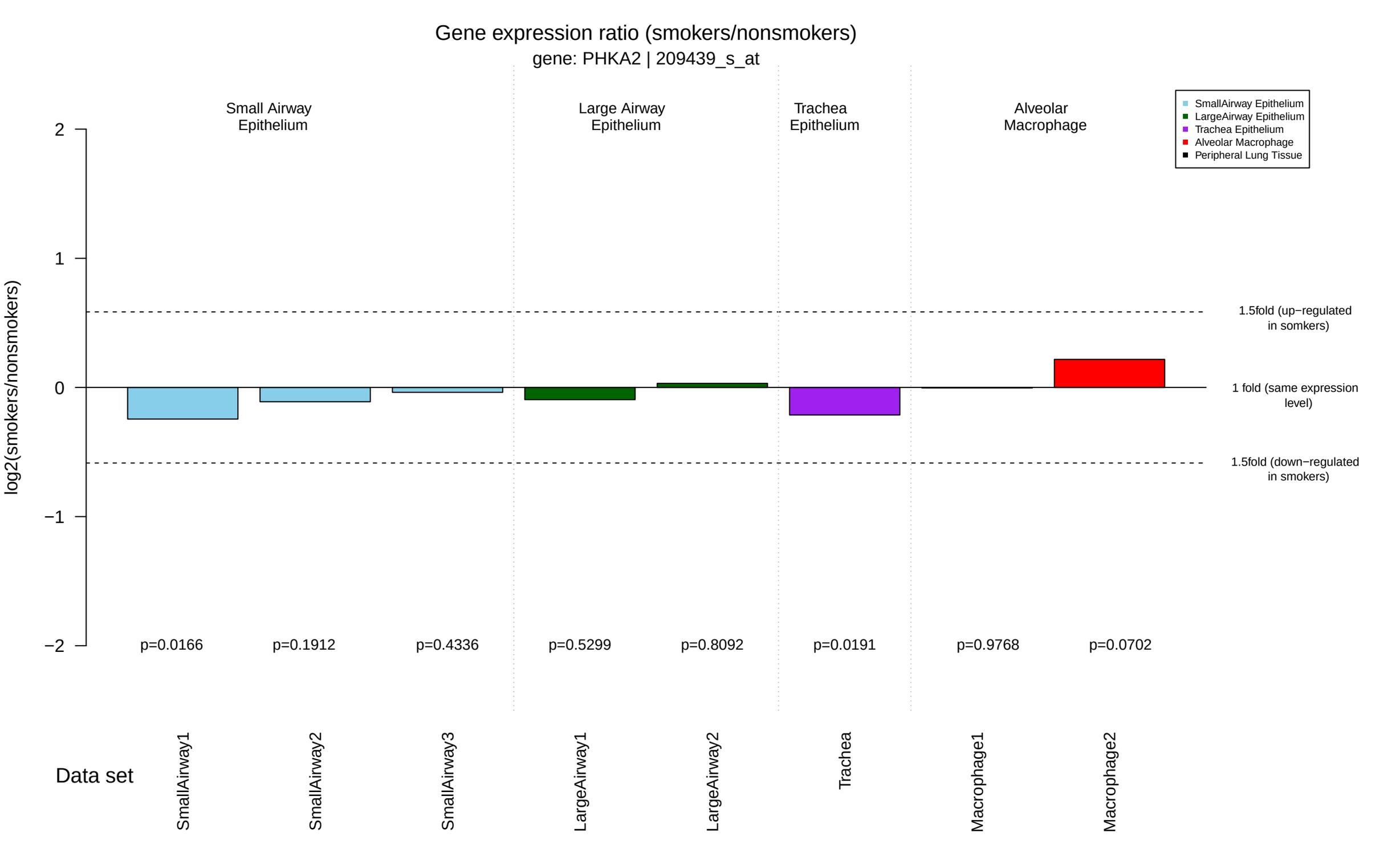
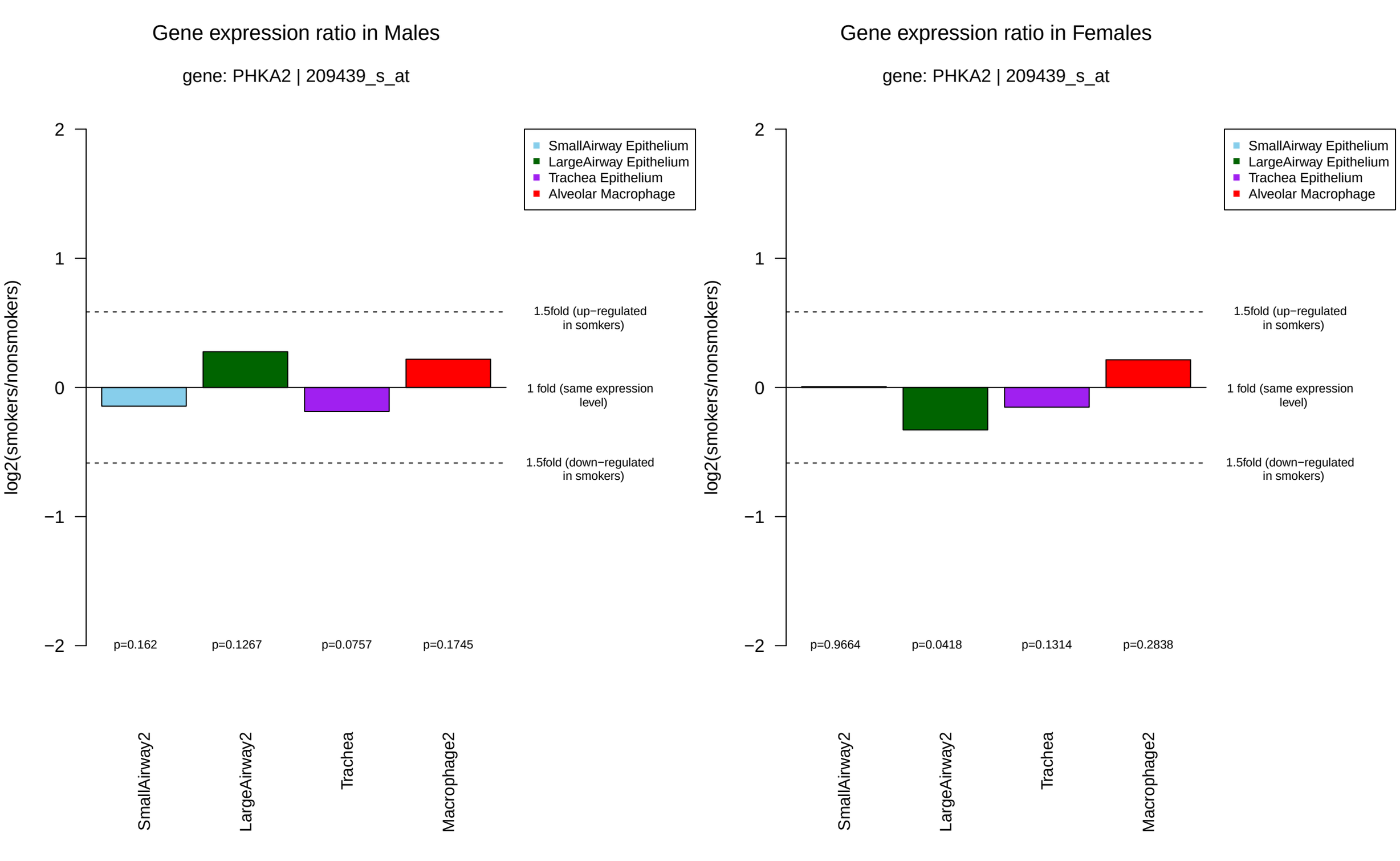
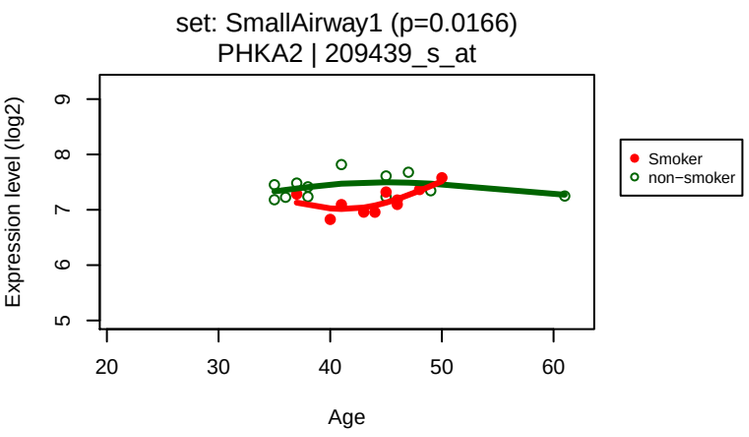
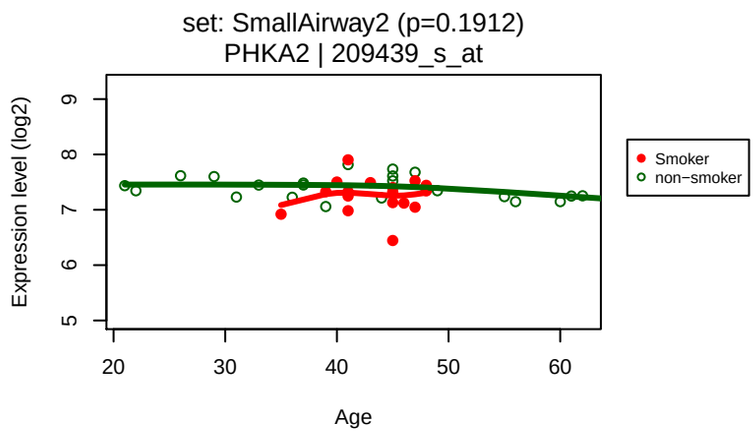
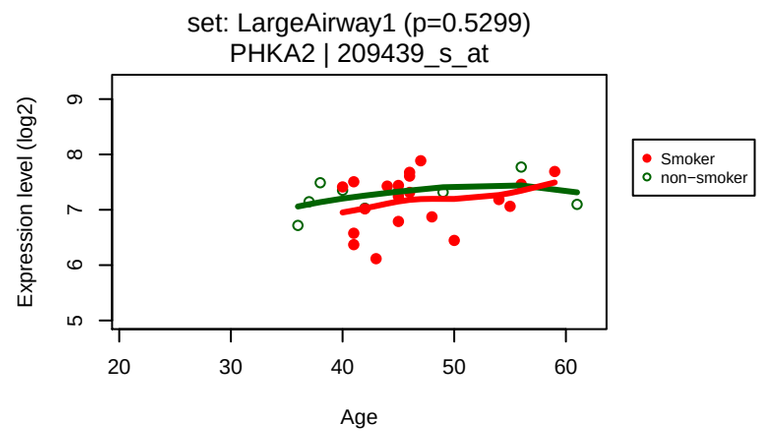

Gene expression of smokers and nonsmokers
gene: PHKA2 | 209439_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0166
p=0.1912
p=0.4336
p=0.5299
p=0.8092
p=0.0191
p=0.9768
p=0.0702
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

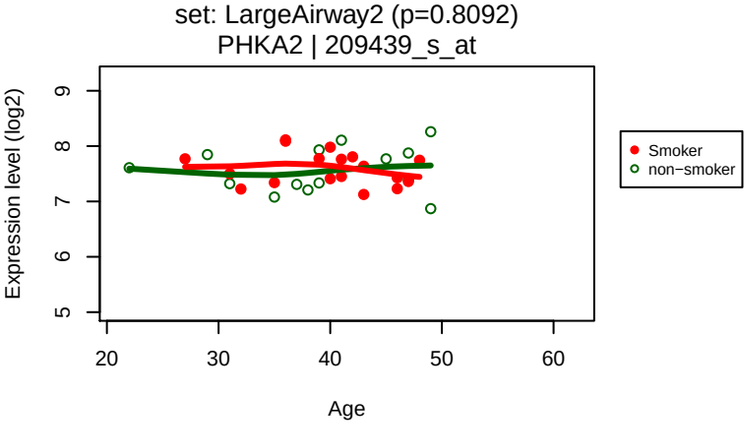

Gene expression ratio (smokers/nonsmokers)
gene: PHKA2 | 209439_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0166
p=0.1912
p=0.4336
p=0.5299
p=0.8092
p=0.0191
p=0.9768
p=0.0702
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

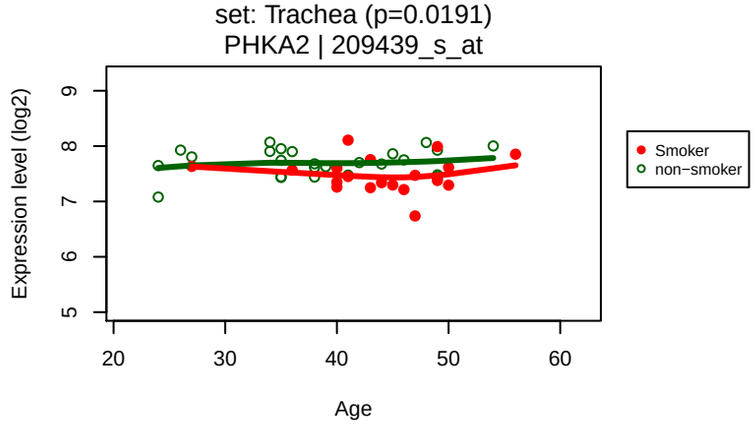

Gene expression ratio in Males
gene: PHKA2 | 209439_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.162
p=0.1267
p=0.0757
p=0.1745
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKA2 | 209439_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9664
p=0.0418
p=0.1314
p=0.2838
SmallAirway2
LargeAirway2
Trachea
Macrophage2

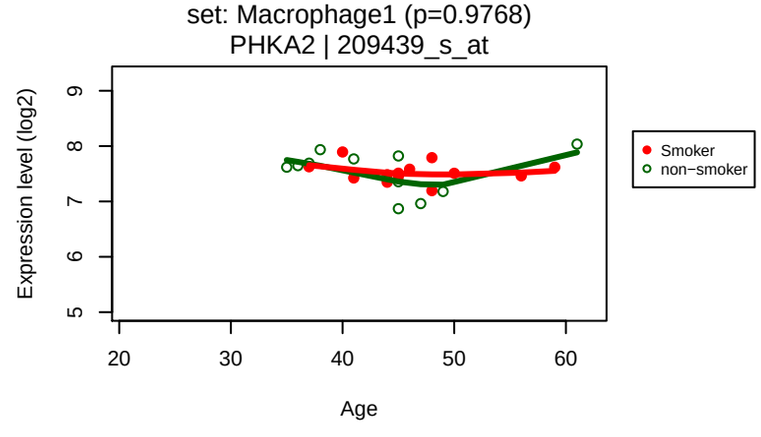

set: SmallAirway1 (p=0.0166)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker

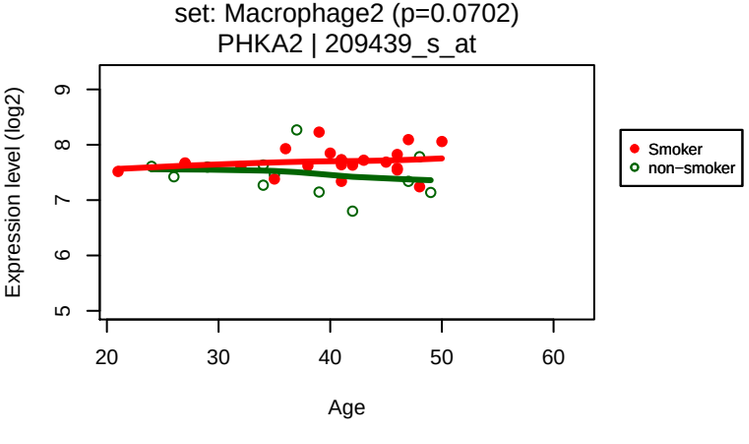

set: SmallAirway2 (p=0.1912)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker

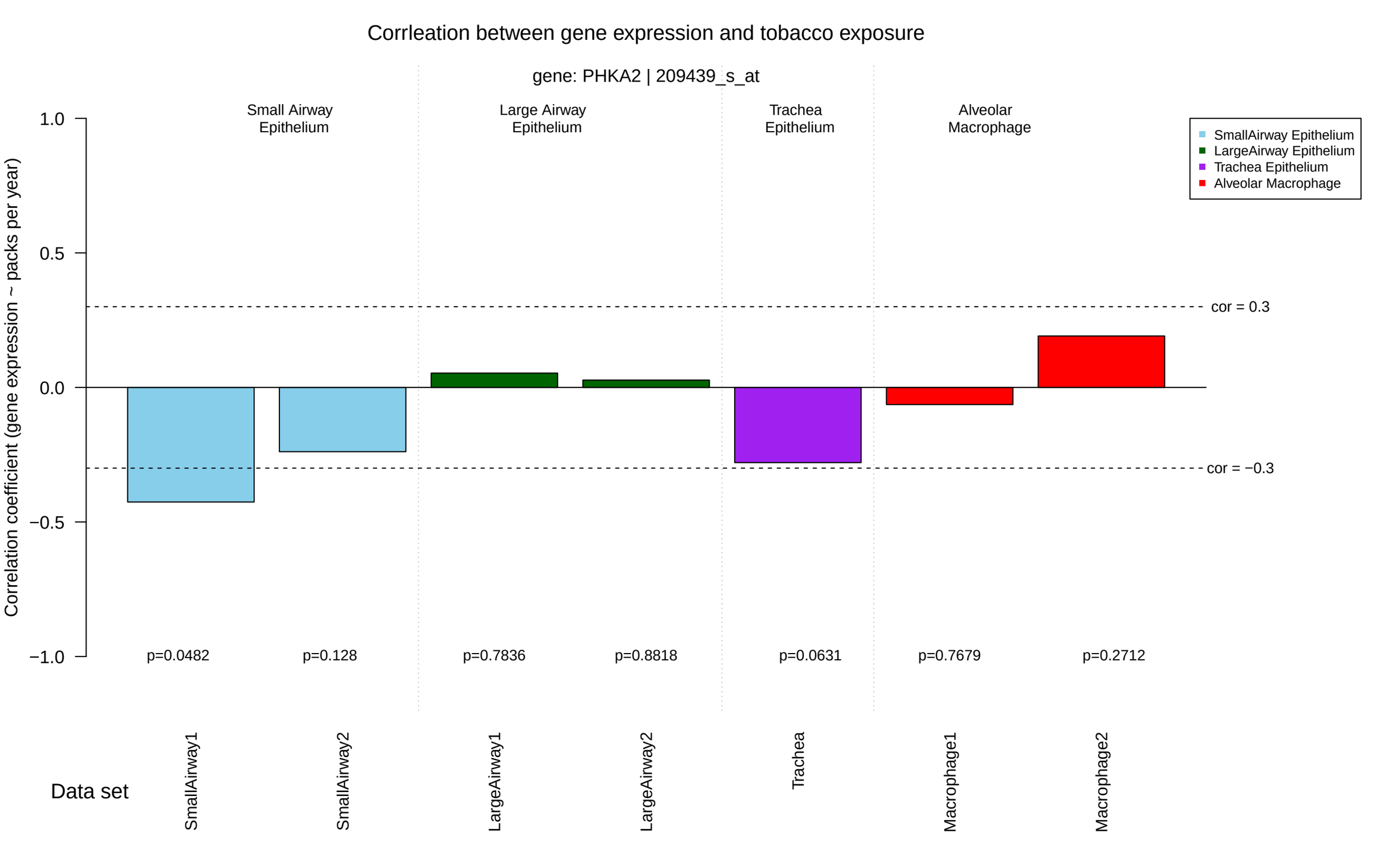

set: LargeAirway1 (p=0.5299)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8092)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0191)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9768)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0702)
PHKA2 | 209439_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKA2 | 209439_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0482
p=0.128
p=0.7836
p=0.8818
p=0.0631
p=0.7679
p=0.2712
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2